

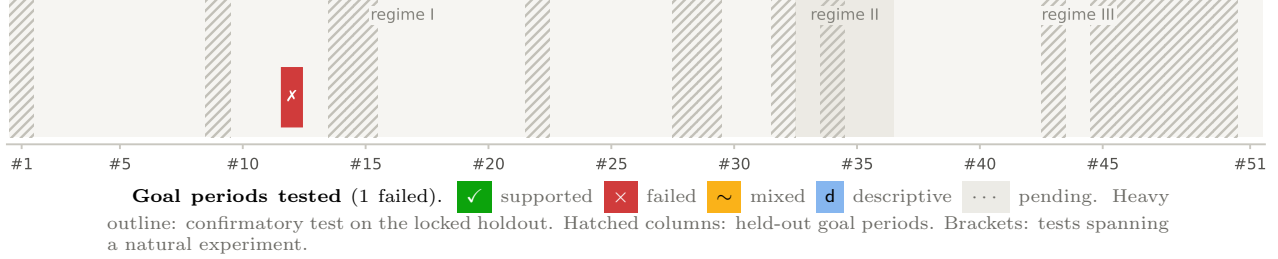
H132: Debate is a two-sublattice limit cycle: topic ping-pong at lag one

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1/5

Bottom line. In #12's debates the team-centred issue projection persists across consecutive turns of opposite teams ($\rho_1 +0.16$, shuffle p 0.002), with no lag-one alternation beyond the turn-shuffled null ($\Lambda +0.05$, p 0.08; power 1.00 at $k_\times = 0.6$).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH375 proposed that a debate is a two-sublattice limit cycle: each team rebuts the other's last move, so the issue content alternates sign at lag one and the cycle runs one way. H132 tests this in #12's ten Asian-Parliamentary debates (306 turns, drafted Government and Opposition teams). Each turn is projected on the motion's support-oppose axis, built from the motion text without labels, and each team's static position is removed. The content does not alternate. Consecutive cross-team turns are *positively* correlated ($\rho_1 = +0.16$, shuffle $p = 0.002$), and the alternation $\Lambda = \rho_2 - \rho_1 = +0.05$ sits inside the turn-shuffled null ($p = 0.08$). The kill fires; the synthetic power for $k_\times = 0.6$ is 1.00. The holdout is not run.

PHYSICAL MODEL

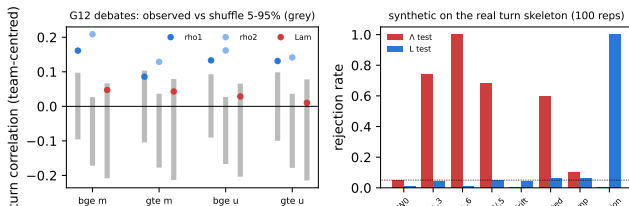
Alternating-update two-sublattice linear dynamics (model 02) with antiferromagnetic block coupling (model 01). y_t is turn t 's whitened projection on the issue axis $\hat{a}_d$; $\epsilon_t = \pm 1$ marks Government or Opposition; μ_d is the assigned (staggered) field; u_t is the dynamic issue state with antiferromagnetic response $k_\times$; z_t removes each team's mean. Speakers alternate by format, so only z_t can carry a cycle.

$$\begin{aligned} y_t &= \epsilon_t \mu_d + a_{i(t)} + u_t + e_t, & u_{t+1} &= -k_\times u_t + \eta_{t+1} \\ \rho_1 &\approx -k_\times r, & \rho_2 &\approx k_\times^2 r, & \Lambda &= \rho_2 - \rho_1 \\ A_\gamma &= \rho_{G \rightarrow O}(1) - \rho_{O \rightarrow G}(1), & L &= \langle a_t g_{t+1} - g_t a_{t+1} \rangle \end{aligned}$$

Observables. Left: observed ρ_1 , ρ_2 and Λ (dots) against the 5–95% band of 5,000 turn shuffles (grey) in four input variants. A cycle needs ρ_1 below its band; it sits above it. Right: rejection rates on the real turn skeleton (100 replicates per world): the Λ test has size 0.05 and power 0.74–1.00 for cycles; drift never triggers it; a rotation is caught only by L .

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator running adversarial or debate formats: assigned sides show up as a static tilt in content, and turn-to-turn content persists across the team line. Expect a rebuttal to restate the point it answers (lag-one correlation $\approx +0.16$ on the issue axis, $+0.15$ in full content), not to swing the topic back and forth. Use stance labels, not embeddings, to see opposition between moves.



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RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1/K $\Lambda > 0, \rho_1 < 0$	$\Lambda +0.047$ (p 0.08); $\rho_1 +0.161$	failed (kill)
P2 $A_\gamma \neq 0$	$+0.048$ (p 0.71)	failed
P3 no within-team alternation	same-team $\rho +0.30$	supported
P4 cycle area $L \neq 0$	-0.071 (p 0.39)	failed
P5 topic echo $\rho_1^{\text{vec}} > 0$	$+0.145$ (p 0.0002)	supported
N1 verdict switch-off	$\Lambda_{\text{post}} -0.18$, no cycle to stop	uninformative
N2 read gate	$\Delta\rho_1 -0.006$ (p 0.47)	failed
N3 re-drafted pairs	-0.047 (18 pairs, p 0.29)	failed

G12 $\times$ failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

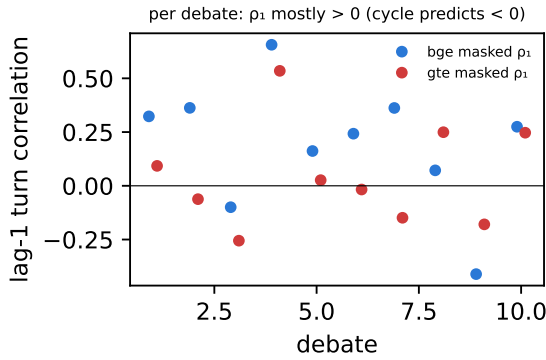
Embeddings encode negation poorly, so a rebuttal that restates a claim reads as agreement on the issue axis; the antiferromagnet H21 and H37 found lives in reply stance. Cycles weaker than $k_\times \approx 0.4$ at H21-sized noise are not excluded (power 0.74 at 0.3). A staggered-field ramp gives false Λ in 10% of synthetic worlds. The negative L in all variants is post hoc. Ten debates, one period, regime I.

ROUND 2

Re-run the turn statistics on stance v2.1 and DQ2 reply stance to the previous turn; fit a common-drift plus team-persistence model; try other alternating formats (#23, #26); run the frozen #34 confirm after sign-off.

- **Essence.** whether assigned opposition creates a dynamic coupling (answer the other side) or only a static field (hold your side).
- **H132-R1** Stance sequence.
- **H132-R2** Drift model.
- **H132-R3** Other alternating formats.

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Second figure. Lag-1 turn correlation per debate in the two masked embedding models. A limit cycle predicts negative values in most debates; $\rho_1 > 0$ in 8/10 debates in bge and 6/10 in gte.